Design Specification — Candidate E2

Selection basis

Candidate E2 was selected because it satisfies all three task targets:

1. Energy density >= 500.94 Wh/kg.
2. 4C fast charge without lithium plating.
3. Maximum temperature <= 60C.

Final design parameters

Parameter	Value	Unit
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Positive electrode thickness	9.828e-05	m
Negative electrode thickness	0.00011076	m
Separator thickness	8e-06	m
Positive electrode porosity	0.35	-
Negative electrode porosity	0.27	-
Positive CC thickness	1e-05	m
Negative CC thickness	8e-06	m
Positive particle radius	3.45e-06	m
Negative particle radius	3.86e-06	m
Heat transfer coefficient	60.0	W.m-2.K-1
Electrolyte diffusivity	8.2e-10	m2.s-1
Electrolyte conductivity	1.35	S.m-1
Cation transference number	0.4	-

Performance summary

Metric	Value	Unit
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Energy density	517.8298883826939	Wh/kg
Volumetric energy density	975.3875141711424	Wh/L
Nominal capacity	6.543863466458118	Ah
Discharge energy	23.544496852671656	Wh
Cell active mass	0.045467628232520006	kg
Active stack thickness	0.00023504000000000003	m
4C max temperature	330.86233607651576	K
4C min anode potential	4.0244993311427946e-05	V

Notes

- The final design uses an architecture-first tuning path within the Chen2020 baseline system.
- Electrolyte density was not included in the mass calculation due to parameter-set limitations.
- The recommended design assumes enhanced cooling relative to the default contract baseline.